

Final Project in Data Analytics

1. General Overview

The final project aims to apply data analytics techniques to solve a real-world problem. Students are required to collect, preprocess, analyze, and interpret data, and present actionable insights using appropriate tools and methodologies.

2.1 Data Collection

- The system/project must utilize at least one **real-world dataset** (e.g., CSV, Excel, API, database).
- Data must contain a minimum of **100,000 records**
- Sources may include:
 - Government open data portals
 - Web scraping (if applicable)

2.2 Data Preprocessing

- Handle missing values (e.g., imputation, removal)
- Remove duplicates and inconsistencies
- Perform data transformation (normalization, encoding, etc.)

Document all preprocessing steps following the **CLEAN Framework** (<https://youtu.be/y9wFFD2bXQM>)

2.3 Exploratory Data Analysis (EDA)

- Generate summary statistics (mean, averages totals)
- Use Appropriate KPI's and measure
- Use visualizations such as:
 - Bar charts
 - Line graphs
 - Other relevant Power BI visuals

1. Use the **PYRAMID FRAMEWORK: KEY METRICS and MEASURES**
<https://youtu.be/9BwpcO-0Lul>

2.4 Data Modeling / Analytics

- Apply at least one analytical method:
 - Use either Star or Snowflake schema for your Data Models

- Descriptive Analytics (trends, summaries)
- Predictive Analytics (Forecasting)

2.5 Visualization & Dashboard

- Develop an interactive dashboard using tools such as:
 - Power BI
- Dashboard must include:
 - Key performance indicators (KPIs)
 - Filters and interactivity
 - Clear and readable charts
- Follow the **DASH FRAMEWORK for your DASHBOARD**
<https://youtu.be/TNfxssxUsTc>

2.6 Insights and Recommendations

- Provide at least **3–5 actionable insights**
- Include data-driven recommendations based on findings
- Interpret results in a real-world context

Groupings: Each Group is composed of 5 Members (With specific Task assigned following the Project Requirements)

RUBRICS: Evaluation Criteria

Criteria	Weight
Data Quality & Preprocessing	20%
Analysis & Modeling	25%
Visualization & Dashboard	20%
Insights & Interpretation	20%
Presentation & Documentation	15%

Published work in Github and documentation using .md format

Output:

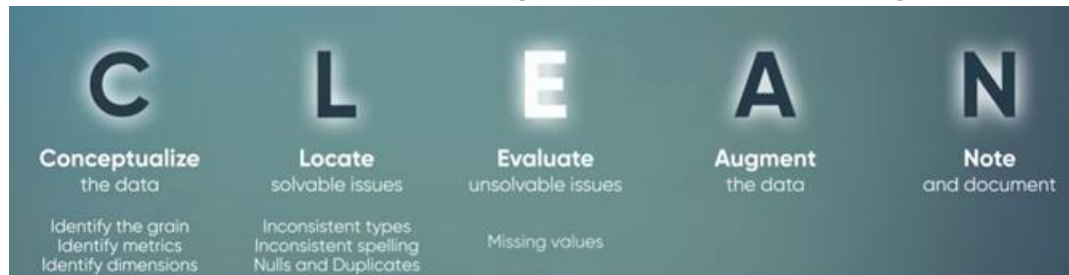
0. PROJECT DESCRIPTION/OVERVIEW

1. Data Collection Procedure

The data set contains 2 data information so that you can compare about the two hotels: a city hotel and a resort hotel

2. Data Cleaning Process / Documentation

Use CLEAN Framework for this task as guide and document cleaning task as shown below



	A	B	C	D	E	F	G
1	Table	Column	Issue	row count	magnitude	solvable?	resolution
2	hotel_Bookings	children	There are missing value/error in the value	4	low	Y	change the error value into 0
3	hotel_Bookings	adult, children, babies	Inconsistent because there are 0 total guest		low	Y	removed if total value is 0
4	hotel_Bookings	all	Duplicates with all the data set		low	Y	remove the duplicates and kept 1
5	hotel_Bookings	agent	missing value (null)	16,340	13.7	Y	change the null to 0 implying no agent
6	hotel_Bookings	company	missing value (null)	112,593	94.3	Y	change the null to 0 implying no company
7	hotel_Bookings	stays_in_weekend_nights, stays_in_week_nights	split variable	119,390	100	Y	combine both into new feature total_stay
8	hotel_Bookings	total_stay	phantom stay or there are 0 stay		low	Y	remove rows where the total stay is 0
9	hotel_Bookings	adult, children, babies	variables are not totaled	119,390	100	Y	combined all clouds into a new named total_guesed

3. Data Model (Star or Snowflake)

Define dim and fact tables with their relationship anchored by the Model

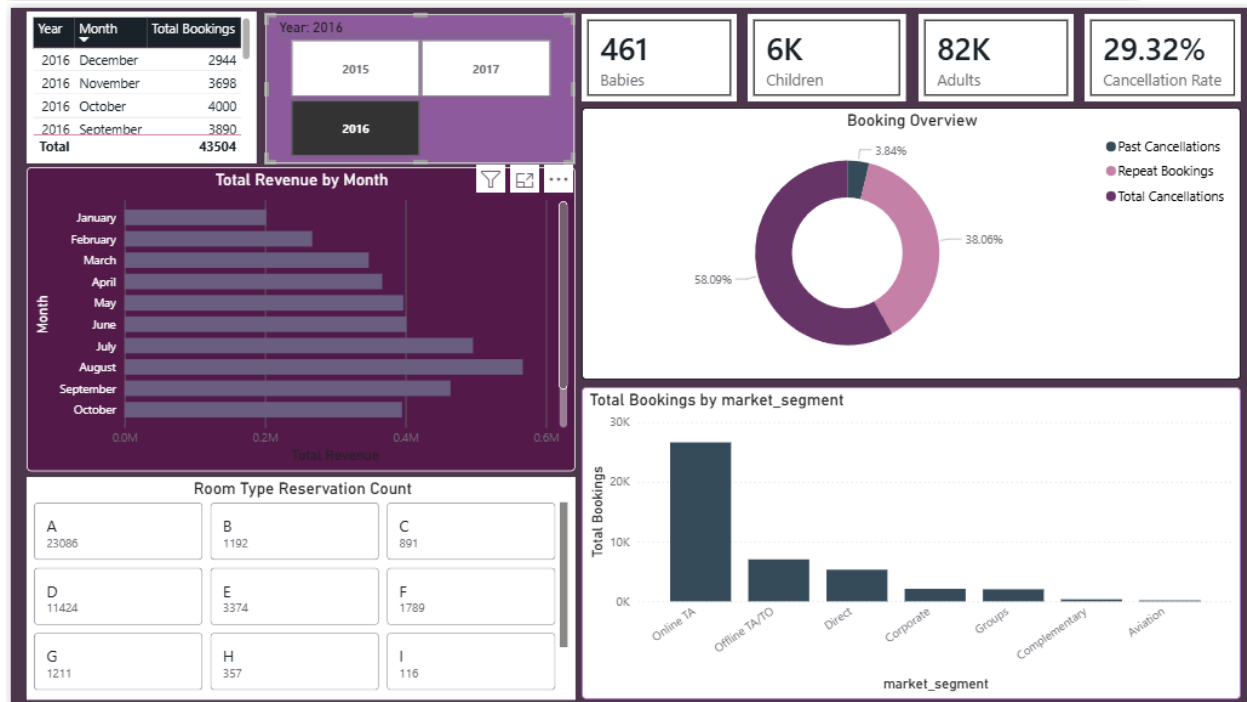
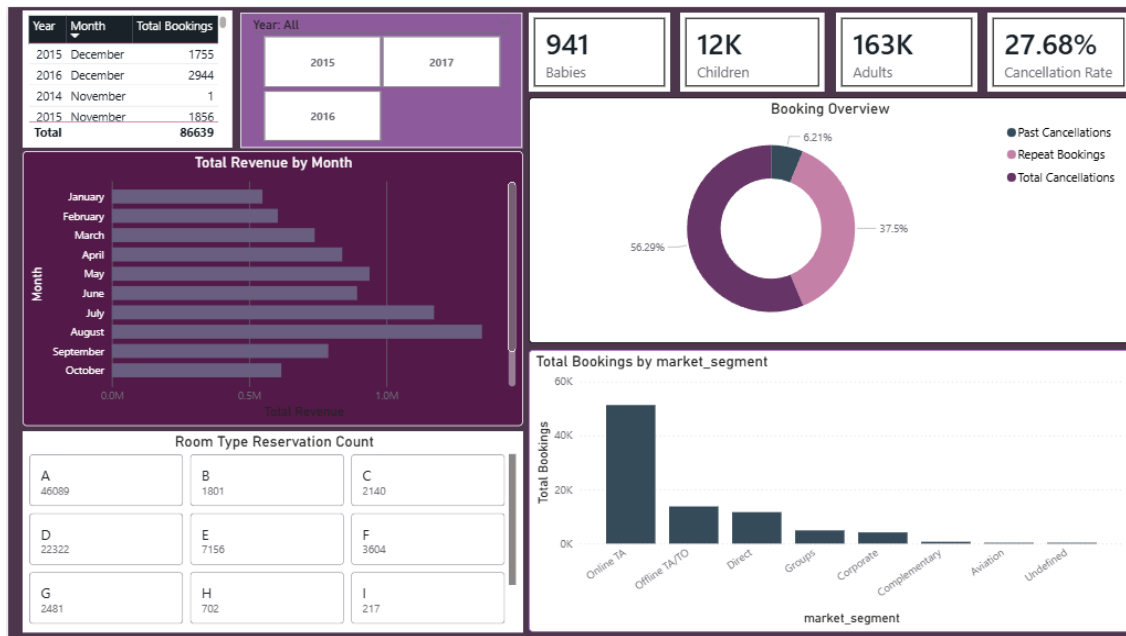
Apply at least one analytical method:

- Use either Star or Snowflake schema for your Data Models
- Descriptive Analytics (trends, summaries)
- Predictive Analytics (Forecasting)

4. Dashboard Wireframe Layout following the DASH Framework (Refer to the finals lecture as guide)

5. Visualization & Dashboard

- **Develop an interactive dashboard using tools such as:**
 - o **Power BI**
- **Dashboard must include:**
 - o **Key performance indicators (KPIs)**
 - o **Filters and interactivity**
 - o **Clear and readable charts**
- **The hotel booking dashboard centralizes booking, demographics, revenue, and cancellations in one interface. It highlights seasonal revenue peaks, room demand, and market segment performance to guide management decisions. Clear metrics on cancellations and repeat bookings support customer retention and strategic planning.**

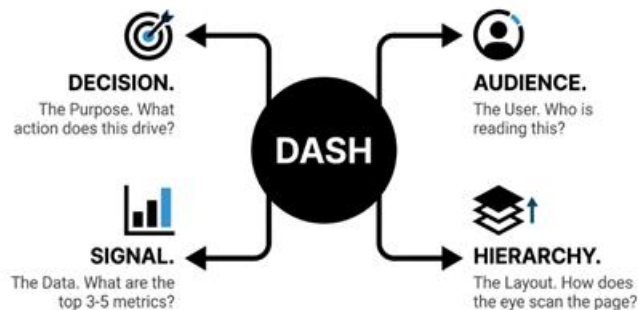


The 2016 dashboard shows bookings were strong, especially in the last quarter. Adults made up most guests, revenue stayed steady with mid-year peaks, and online travel agencies drove the majority of reservations. Room Type A was the top choice. The main issue is the high cancellation rate, but repeat bookings are solid, showing guests do come back. Overall, performance looks good, but reducing cancellations would make it even better.

Follow the DASH FRAMEWORK for your DASHBOARD

<https://youtu.be/TNfxssxUsTc>

The Four Pillars of the DASH Framework



6. Insights and Recommendations

a. Provide at least 3–5 actionable insights

***Most customers prefer booking through third-party travel platforms because they are convenient and widely accessible.**

***Most bookings are adult travelers, likely couples, business travelers, or solo tourists.**

***This room type likely offers the best balance between price, comfort, and features.**

b. Include data-driven recommendations based on findings

***Improve the hotel's website and mobile booking experience to encourage direct reservations.**

***Focus marketing on adult-friendly packages such as business stays, romantic getaways, or wellness packages.**

***Analyze why customers prefer Room Type D and replicate those features in other room categories.**

c. Interpret results in a real-world context

***Since customers book through third party travel platforms the hotel can be seen but they usually charge commission fees, which reduce the hotel's earnings per booking.**

***The hotel mainly serves: couples, business travelers, solo tourist and adult leisure travel**

***This room type must be offering the best value for their money. It may have better amenities or more affordable pricing**

